

SPRINT 2026

Project Description

PROJECT NAME: STREET VIEW ROBOT

MENTOR: MR. RAJAEI KHATIB

GOAL:

This project seeks to design a navigation robot that traverses a room while constructing 360° panoramic and point cloud representations of its surroundings at each sampled location.

DESCRIPTION:

This project presents a mobile robot system capable of traversing a room while constructing 3D point cloud representations of its surroundings at each sampled location.

The robot is built on the RasbotV2 platform, a Raspberry Pi-based chassis equipped with four Mecanum wheels that enable omnidirectional movement -- forward, backward, lateral strafing, diagonal, and in-place rotation -- all controlled through an I2C motor driver.

An Intel RealSense D405, which contains a factory-calibrated stereo infrared pair, is fixed to the chassis for depth sensing. Additional sensors include an ultrasonic rangefinder for obstacle distance measurement and four line-tracking sensors for ground reference.

Before beginning, each group must assemble the robot chassis, mount the motors, wheels, camera, and sensors, and initialize the Raspberry Pi -- installing the operating system, configuring I2C and camera interfaces, and setting up the Python environment with the required libraries (OpenCV, NumPy, pyrealsense2).

At each sampled location, the robot rotates and captures stereo frames. The students must design and implement an algorithm to compute depth from the stereo pairs, generate 3D point clouds, and merge the views from different rotation positions into a single 360-degree point cloud of the surroundings.

The entire system is built as a Python API that abstracts all hardware into simple method calls -- `forward(speed)`, `capture_stereo()`, `set_tilt(angle)` -- and supports two navigation modes:

Mode 1: Manual Control

The robot is operated interactively through a keyboard-driven terminal interface. The user drives the robot in real time using WASD keys for omnidirectional movement and in-place rotation, navigating it to desired locations within the room. At each location, pressing a single key triggers the point cloud capture routine, which runs autonomously (rotating, capturing, and merging) before returning control to the user.

Mode 2: Autonomous Line Following

A dark tape path is laid on the floor connecting the desired capture locations. The robot follows the line autonomously using its four infrared line-tracking sensors. Stop markers -- perpendicular cross-marks on the tape that trigger all four sensors simultaneously -- indicate capture locations. When the robot detects a stop marker, it halts, performs the point cloud capture routine, and then resumes line following to the next marker.

Both modes allow the user to incrementally build a collection of point clouds across the room.